



# **International Crude Oil Markets:** Benchmarks, Logistics, Refining Margins & Hedging



## International Crude Oil Markets: Benchmarks, Logistics, Refining Margins & Hedging

Crude Oil and natural gas lie at the heart of the global energy economy. Their journey—from upstream extraction to processing, transportation, and final market delivery—is shaped by geopolitical dynamics, logistics infrastructure, commercial terms, and the strategic behaviors of producers and traders. This course offers an integrated view of the international oil and natural gas market structures, covering the supply and demand forces, transportation logistics, refining and processing economics, and trading strategies that define value creation. Participants will develop a practical understanding of how market signals, contracts, and physical flows interact across borders—supporting more effective decision making within investment and risk management domains.



### LOCATION

Rio de Janeiro

London

Singapore

**LONDON PETRO** ACADEMY  
Empowering Professionals for a Sustainable Future



# COURSE OBJECTIVES

## By the end of the course, participants will be able to:

- Identify major crude oil benchmarks and global supply routes.
- Understand the regional natural gas markets and the Global liquified natural gas
- Describe refining processes and how crude quality impacts yields.
- Analyze trading contracts for managing price risks.
- Evaluate geopolitical events affecting oil and gas supply and market prices.
- Develop a hedging plan using futures or swaps.
- Calculate refining margins using crack spread models.
- Assess arbitrage between crude grades based on cost and yield.
- Develop an economic framework based on market understanding for investment and project appraisals.

## Who should attend:

- Energy professionals in commercial, trading, or strategy functions
- Decision makers in supply chain, logistics, or refinery planning
- Financial analysts, risk managers, and project evaluators
- Energy analysts & project managers
- Petroleum engineers, Facility Engineers
- Geology and Geophysics professionals
- Business development & corporate strategists
- Analysts and market experts
- Legal and regulatory professionals engaged in trading or transportation contracts
- Researchers and stakeholders seeking a comprehensive understanding of global crude markets

## Training Methodology

This course provides comprehensive learning experience that integrates the instructor's firsthand experience in international energy companies with a robust theoretical framework. Drawing from years of practical experience in the oil & gas sector, the instructor brings real-world insights and industry best practices to the discussions.





# Day 1

## Introduction to Global Oil and Regional Gas Markets

- Overview of the crude oil value chain: upstream to downstream
- Overview of natural gas value chain
- Fundamentals of oil and gas industry: geoscience and engineering
- Market fundamentals: supply sources, demand centers, and trade flows
- Major producers, consumers, and OPEC+ market influence
- Project economics: cost structures and value drivers

# Day 2

## Transportation and Logistics

- Energy Commodity Price Analysis
- Crude oil transportation: pipelines, tankers, and maritime routes
- Regional Natural Gas Markets
- Oil price formula: time, location, and grade differentials
- Storage dynamics and contango/backwardation implications
- Case study: Economics of transporting heavy vs. light crudes

# Day 3

## Downstream Economics

- Refinery configurations and complexities
- Oil benchmarks: Brent, WTI, Dubai, and price-setting dynamics
- Crude assay: gravity, sulfur, and compatibility with refineries
- Refining margins and crack spreads
- Natural gas trading and optimization
- Liquefied Natural Gas: a global market
- Market instruments: Forwards, Futures, and Options
- Arbitrage opportunities between crude grades and refining destinations





## Day 4

### Trading Strategies and Risk Management

- Participants in physical and paper energy commodity markets
- Risk and volatility in prices
- Energy commodity contracts: spot, term, and long-term offtake agreements
- Hedging strategies for producers, refiners, and traders
- Is perfect hedge possible? Strategies for risk management
- Price risk, basis risk, and geopolitical uncertainty
- Case study: setting up a hedge for a hypothetical cargo

## Day 5

### Geopolitics, Emerging Trends, and Market Outlook

- Geopolitical tensions and sanctions impacts
- Shifts in refining capacity and global crude flows
- Gas pipeline geopolitics
- Environmental regulations, ESG pressures
- The role of shale, biofuels, and alternative blends
- Future trends: digital trading, emission markets, and maritime tracking
- Capstone group exercise: advising a trading desk on sourcing and hedging strategies







# BABAK JAFARIZADEH

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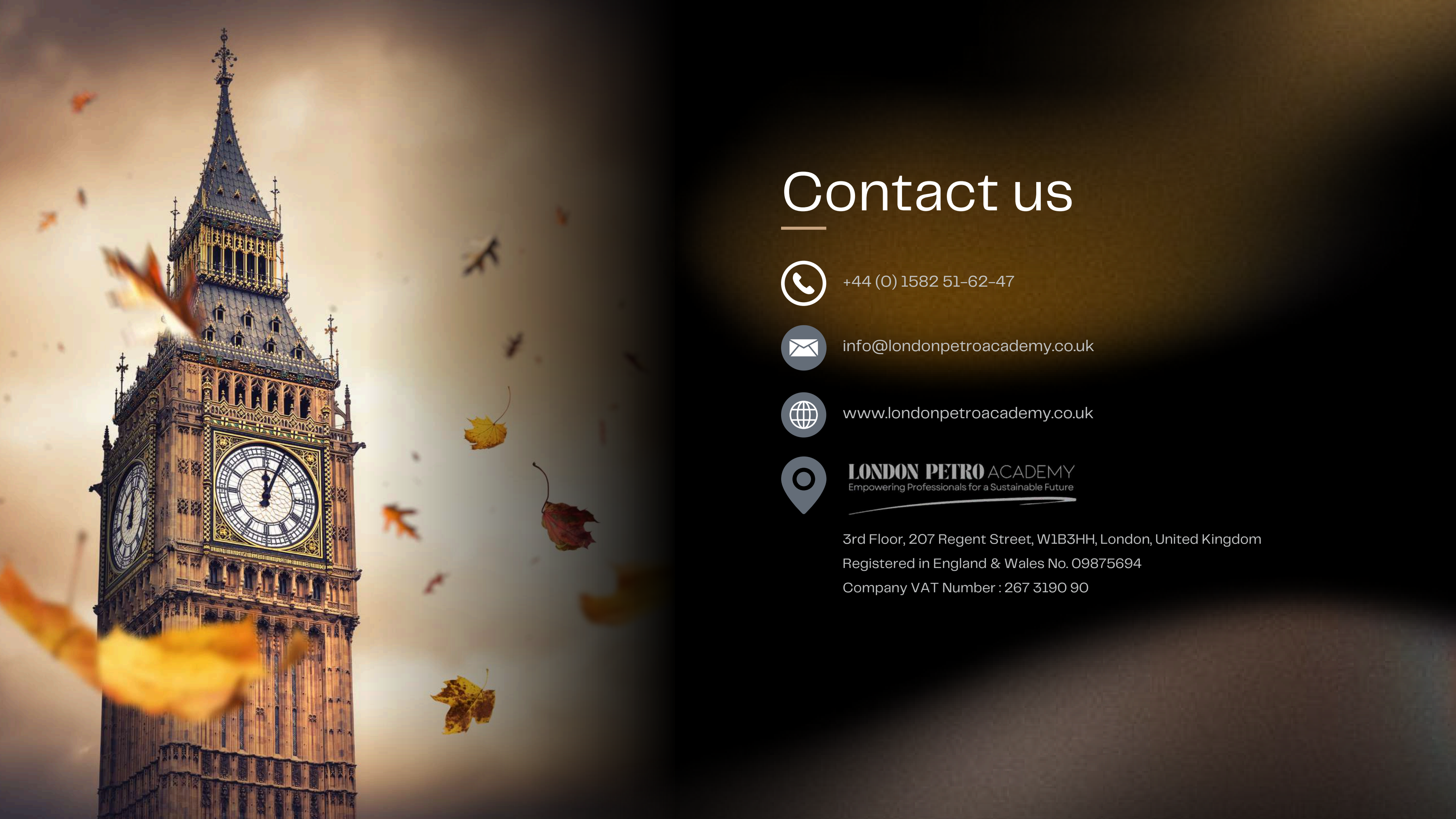
Dr. Babak Jafarizadeh is a senior consultant and engineering economist who views engineering as the pursuit of economic solutions to technical challenges in an uncertain world. With extensive experience in the oil and gas industry, he has served as a senior economic analyst supporting multi-million-dollar investment decisions for projects across the globe.

Since 2019, Dr. Jafarizadeh has worked as a senior consultant for leading training and consulting companies, while also holding academic appointments at several prestigious institutions — including Heriot-Watt University in the UK (Assistant Professor), the University of California, Berkeley (Lecturer at the Fung Institute for Engineering Leadership), and previously the University of Greenwich (Senior Lecturer). His earlier industry experience includes serving as a Senior Economic Analyst with Equinor (formerly Statoil) in Norway and the USA.

Dr. Jafarizadeh holds a PhD in Petroleum Engineering with a concentration in Investment and Decision Analysis from the University of Stavanger (in collaboration with the University of Texas at Austin), as well as an MSc in Financial Engineering and a BSc in Industrial Engineering. He is an active member of several professional associations, including the Society of Petroleum Engineers (SPE), the Institute for Operations Research and the Management Sciences (INFORMS), and the European Association of Geoscientists and Engineers (EAGE).

His teaching and research interests lie at the intersection of decision analysis and corporate finance, with particular applications in engineering and the energy industry. He has designed and delivered numerous professional training courses, such as Advanced Financial Modelling for Oil and Gas Professionals, Advanced Project Economics, Oil and Gas: Markets, Trading, Pricing and Economic Frameworks, and Energy Economics. Dr. Jafarizadeh is also the author of the book Economic Decision Analysis (for Project Feasibility Studies) and continues to contribute to both academia and industry through research, consulting, and executive education. His areas of expertise include petroleum economics, investment and decision analysis, project feasibility studies, corporate finance, and financial modelling for energy projects.





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